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Managing Fatigue in Team Sports: A Brief Review of Concurrent Training Effects Within the Microcycle

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ABSTRACT

Concurrent training (CT), which combines resistance exercise and energy systems conditioning, is the default approach to preparation in high-intensity intermittent ("stop and go") team sports. This review provides an overview of CT, emphasizing its complexities and challenges in managing fatigue and optimizing performance. These complexities are specifically compounded by the variability in game demands across the season, where the presence of intensified and nonintensified competition periods necessitates a flexible and adaptive training approach. In this context, there are essential training variables to consider, including intensity, volume, session order, and recovery intervals between sessions. In addition, nontraining variables such as travel, sleep, and nutrition play a role in the fatigue experienced while training and competing. These variables interact to influence acute performance and training adaptations and can be strategically adjusted by strength and conditioning practitioners. The aim of this review is to provide a comprehensive understanding of fatigue management for practitioners in team sports, emphasizing the complexities and challenges of CT and offering simplified practical recommendations for adjusting training variables within any given microcycle.

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INTRODUCTION

High-intensity intermittent team sports, such as soccer, basketball, handball, and others, alternatively called "stop and go" sports, exhibit a high degree of complexity. These sports are usually characterized by unpredictable patterns during matches with frequent transitions between brief, multidirectional, high-intensity movements and extended periods of low-intensity activities (1). Because of the intricate nature of these sports, athletes are often required to develop concurrently diverse physiological attributes, including, but not limited to, cardiorespiratory fitness and strength/power (5). Given the wide spectrum of physical capabilities required, it is common practice for team sport athletes to engage in training regimes that integrate both resistance and energy systems conditioning (i.e., continuous endurance exercise, high-intensity interval training [HIIT], sprint interval training [SIT]), which is a practice known as concurrent training (CT) (5). Although CT is a necessary component to prepare athletes for the physiological demands of most team sports, it is essential to acknowledge that its implementation presents logistical complexities, particularly in high-performance sport settings (5,7).

High-performance athletes participating in these sports, notably those engaged in professional leagues, are consistently subjected to not only heightened training load but also to substantial competitive demands (6). These elevated demands are the

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consequence of various factors, such as a heightened frequency of domestic competitions and the intensified competitive nature driven by advanced player strategies (3). Because of the highly congested training/competition schedule these athletes encounter, fatigue and recovery may limit performance and the team's success (Figure 1) (6).

Indeed, fatigue management is pivotal in performance enhancement and injury reduction (4). The objective is not only to minimize the risk of injuries and/or overtraining but also to avoid the pitfalls of undertraining, a concept referred to as the "sweet spot" (33). An undertraining state can also elevate injury risk because of poor fitness levels (33). However, the intricacy of fatigue management is amplified when athletes participate in CT regimens (2), which may further exacerbate performance stagnation and injury risk because of the increased training volume or the acute training stress (5). Therefore, because fatigue management is a critical component for team success (4), it becomes imperative to formulate strategies to mitigate the adverse effects of CT programs, thus ensuring the sustained maintenance of optimal performance levels. Current guidelines extensively address strategies for mitigating the interference effect (10,11), but there is limited consideration of the dynamic fluctuations in training load because of periodization. Consequently, existing recommendations often overlook the nuanced adjustments required to optimize performance across varying intensities of work.

Fatigue management in team sports has been extensively researched, yet the practical integration of CT variables within a dynamic and applied framework remains underexplored. Unlike systematic reviews, such as Silva et al. (94) and Sousa et al. (97), which provide focused examinations of

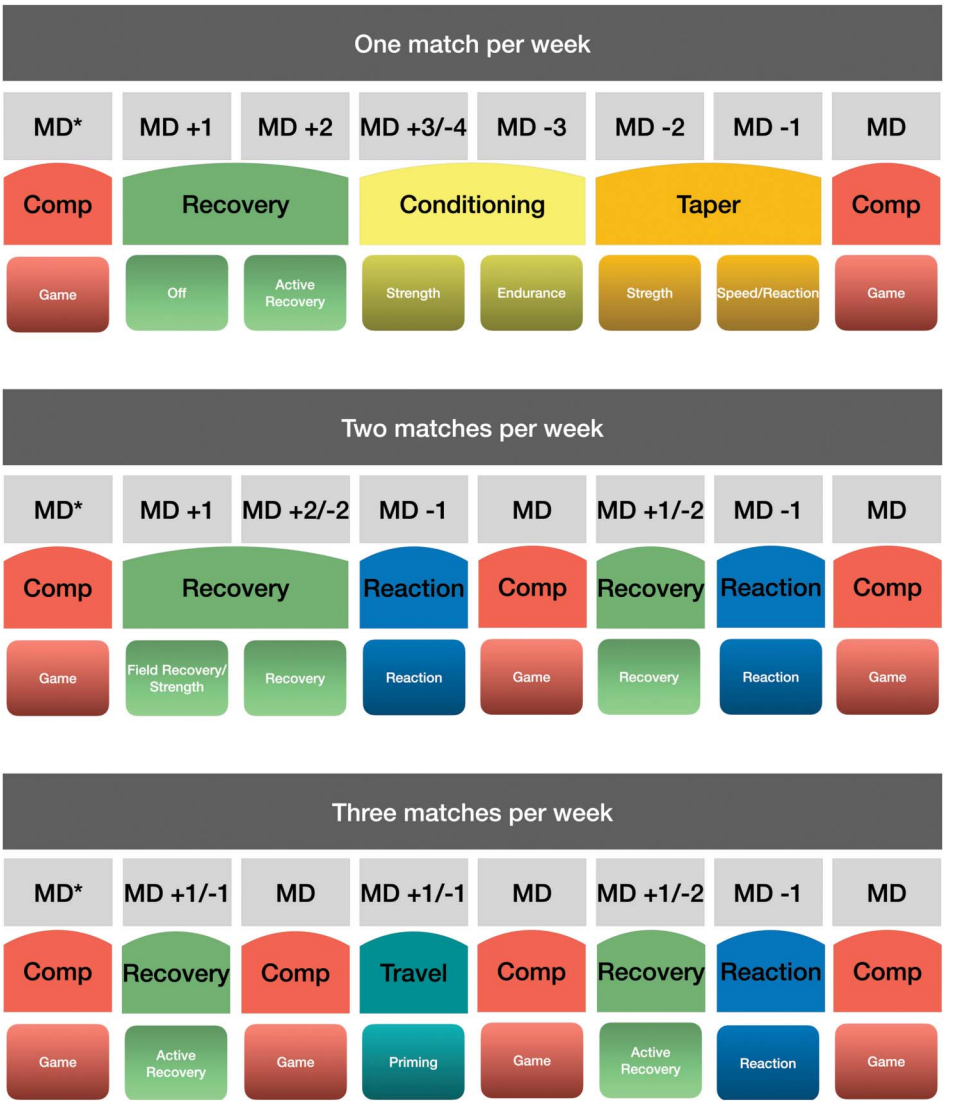

Figure 1. Schematic representation of concurrent training demands during different periods of fixture congestion. Comp = competition; MD = match day.

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pitch-based training loads and exercise intensities, our narrative review is designed to offer evidence-based, adaptable guidelines for strength and conditioning practitioners. Specifically, we address the cumulative effects of training variables, such as intensity, volume, and session order, while incorporating underrepresented nontraining factors like travel, sleep, and nutrition. Recent works have provided practitioners with updated programming guidelines (22), periodization strategies (70), and monitoring approaches (19), which can be used prescriptively. This review complements these efforts by suggesting a decision-making approach that allows practitioners to dynamically integrate and adjust multiple variables. By synthesizing recent findings and providing practical tools, the aim of this review is to offer an adaptable approach to fatigue management that aligns with the real-world demands of high-intensity intermittent team sports.

CONCURRENT TRAINING INFLUENCE ON FATIGUE AND INTERFERENCE EFFECT

CT has been defined as the simultaneous incorporation of resistance and aerobic training within a periodized training regime (4). Although many studies on CT typically incorporate low-intensity continuous aerobic exercise to control for fatigue (5,8), this approach is not commonly observed in team sports settings (2,10,11). In the context of team sports, practical considerations such as the nature of the sports and limited training time have led practitioners to primarily use small-sided games (SSG) (2,11) to address energy systems conditioning alongside technical and tactical aspects. Alternatively, HIIT and SIT are preferred over long continuous low-intensity aerobic training (10,11). This happens because HIIT and SIT involve short bouts of high or maximal intensity work followed by rest periods designed to match or exceed the metabolic demands of the sport (61). Consequently, for this article, we will use the term "energy systems conditioning" instead of "aerobic training" to encompass SSG and HIIT and SIT as methods for metabolic and neuromuscular system conditioning.

Although CT is a prevalent practice in team sports, a primary concern with this approach is the possibility that energy systems conditioning may compromise the adaptations induced by resistance training (i.e., enhanced maximal strength, power, or rate of force development) (1,4). Conversely, evidence suggests that resistance training can further enhance endurance performance by enhancing movement economy (8). It is worth noting that although long-term improvements in endurance performance do not seem to be hindered by strength training (4), the potential exists for residual effects arising from strength sessions to exert adverse influence upon subsequent energy systems conditioning sessions (3). This impact may become evident in the form of disruptions in running technique and a decline in overall biomechanical effectiveness during a subsequent session related to the fatigue of the athlete (3). This becomes a particularly important consideration for team sports athletes during SSG as this may contribute to an increased risk of injury; therefore, practitioners should be cautious about these negative implications.

The term "interference effect" was originally coined by Hickson in 1980 (43) to describe the phenomenon where concurrent strength and endurance training leads to smaller strength gains compared with strength training alone. Despite published evidence for the interference effect, the underlying mechanisms are yet to be fully elucidated (6). In practical settings, concerns that CT might compromise performance are largely attributed to the acute interference hypothesis (6). According to this hypothesis, it is suggested that a previous training session can have a substantial negative impact (induce fatigue) on the "quality" of the subsequent training session (7,9). In this context, "quality" refers to the athlete's ability to meet the prescribed intensity and volume of training, which may lead to suboptimal training adaptations and an increased injury susceptibility (5,9). Therefore, coaches must develop a well-structured strategy to minimize these negative events. This strategy should be tailored to effectively mitigate these outcomes while considering the intricate logistical challenges that practitioners and athletes encounter in their training environments.

In the context of CT, there are some essential training factors to consider. These include elements of both energy systems and strength training, such as intensity, volume, and/or session order. Practitioners need to have an operational understanding of how these training variables interact to optimize performance. Considering energy systems training is more likely to be specific to the preparation of various team sports (61), the complexity of training variable manipulation cannot be specifically addressed for all scenarios. However, the prescription of resistance training (RT) for team sports tends to be more general, where the purpose is to enhance general force production and reduce injury risk (99) and not necessarily the pursuit of lifting the heaviest possible external loads. Thus, the balance of this review will provide a greater quantity of information on the effects of RT variables on acute fatigue and, therefore, CT, reflecting the current state of the literature. In addition, nontraining variables such as travel, sleep, and nutrition can be strategically adjusted. When prescribing training, practitioners can manipulate these training and nontraining variables to limit the interference effect as much as possible and reduce injury risk.

TRAINING FACTOR CONSIDERATION

Intensity. Manipulation of training intensity to elicit favorable adaptations to both resistance and endurance type exercise has long been established (106). Regarding RT, intensity is commonly defined as relative load (e.g., percentage of 1 repetition maximum

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[1RM]), which is used to prescribe training loads to provoke a specific adaptation. High-intensity loads (>80% 1RM or < 8RM) seem to be minimum thresholds to evoke changes in maximum strength (91). The range of intensities required for muscle hypertrophy, the other key outcome for RT, is much wider, as low-, moderate-, and high-intensity RT seem to stimulate similar hypertrophy once the total training volume is adequate, and the number of repetitions performed per set approaches muscular failure (69).

Despite the clear crossover in outcomes, proximity to failure does not seem to be as important in developing strength and power as it does for developing muscle size, especially at lower training volumes (36). In addition, avoiding failure may be favorable when strength is the main outcome of interest (36,59) and is not detrimental to muscle hypertrophy (36,84). Given that resistance exercise performed to failure seems to increase recovery time because of greater muscle damage and neuromuscular fatigue compared with nonfailure training (104), avoiding failure while keeping training intensity high is likely to yield sufficient strength gains (36), which should reduce the likelihood of compromising subsequent training sessions, especially when training and competition periods are congested.

Strategies such as rest-pause and cluster training have been offered as viable alternatives to traditional resistance exercise for long-term increases in strength and rate of force development (37). These strategies manipulate intra-set rest to maintain higher velocities during those repetitions that occur later in the set, where movement velocity would normally decrease because of fatigue during a traditional set (72,101,102). Although the within-set fatigue seems to be reduced by these methods, the impact that such strategies may have on subsequent training sessions is yet to be described in the literature (60). However, higher velocity movements have been demonstrated to increase recovery timeframes after resistance exercise (45). This effectively results in an artificial increase in intensity on a per-set basis, which may be beneficial if adequate recovery is available; however, the likely small gain in additional stimulus may not be proportional to additional recovery costs incurred. Indeed, Nájera-Ferrer et al. (71) recently demonstrated that a higher velocity loss (VL) threshold during a straight set resistance exercise session resulted in greater metabolic stress and, subsequently, increased ventilatory cost and reduced running distance during subsequent low-intensity aerobic endurance exercise. Because of the inconsistencies in outcomes when rest-pause techniques, such as clusters, have been compared with traditional set structures over the longer term (53), further research is needed, and practitioners should be cautious when using this method in the context of a CT microcycle.

The manipulation of training intensity for developing aerobic/anaerobic energy system conditioning is more variable than RT because of the various modes and defined intensity thresholds and concepts, such as $\dot{V}O_2max$, lactate threshold, and critical power (14). Repeated "all out" efforts, the highest level of intensity in the "endurance" type exercise continuum, considering the supramaximal relative speeds/loads, that is, those speeds beyond a minimal speed threshold that elicits $\dot{V}O_2max$ up to and including maximal sprinting speed (MSS) (12,88). Interestingly, both central and peripheral fatigue seem to take between 48 and 72 hours to recover, in a similar manner to heavy (>80% 1RM) resistance exercise (100). Exposure to high-speed running is considered an important component in the physical preparation of intermittent field sport athletes, not only to condition the metabolic systems to meet the high intensity demands of each sport (14) but to play a role in reducing injury risk (65) and maintain or increase running velocity (9).

Surprisingly, the effects of various RT intensities on subsequent conditioning or sports-specific training sessions have not been directly investigated. However, given the recovery demands of high-intensity RT (100), its strategic inclusion within a microcycle warrants careful planning to avoid potential interference with subsequent sessions. It is not yet possible to derive the optimal recovery time between high intensity bouts, although some studies have suggested between 6 and 24 hours as a starting point (83), but this is heavily dependent on the current work capacity of the athletes (49). More studies are needed, and these designs should consider the nature of the upcoming energy systems training session, as interference tends to be measured during an incremental test and/or time to exhaustion test (24). This is sufficient to demonstrate the effect but may not reflect the true demands of an upcoming training session that could include a skill or cognitive component or resistance exercise with high external load and/or volume.

Volume. Resistance training volume is commonly calculated by multiplying the number of sets by the number of repetitions to estimate volume load (55). Other methods to calculate volume also exist, including counting the weekly sets per muscle group/exercise that approached or reached voluntary failure (92). Some authors have sought to determine a RT volume that is optimal for strength and hypertrophy gains (8,79,80). Because of discrepancies in estimation of training volume, proximity to failure, and exercise selections between studies, it is difficult to determine narrow volume landmarks for strength and hypertrophy. Despite this, the available data suggest that higher training volumes are favorable for muscle mass increases regardless of intensity, yet maximal strength increases require higher intensities but are not necessarily volume dependent (90). Interestingly, strength gains can be achieved by using volumes as low as a single set per session performed 3 times per week, provided they are

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performed at high intensity (load > 70% 1RM) and a high perception of effort based on a meta-analysis by Androulakis-Korakakis et al. (2). Higher set volumes between 2 and 6 sets per exercise within a session can offer greater strength gains (47,56), yet the magnitude of difference may not be meaningful to team sport athletes as they do not necessarily need to chase maximal adaptation but only sufficient adaptation. Because adaptation is the result of training stress and disruption to muscle homeostasis, some fatigue is required and expected (96). Therefore, in practice, the goal is simply to avoid those high training volumes that may be impractical or unnecessary and may indeed contribute to accumulated fatigue in the context of training and competing (54).

The dose-response effects of resistance exercise volumes on markers of muscle damage, neuromuscular fatigue, metabolic fatigue, and subsequent training performance have not been explicitly explored by the literature. However, higher training volumes increase the likelihood of compounding accumulated fatigue (22,54). It stands to reason that higher session volumes would exacerbate known acute and delayed responses to resistance exercise, such as increased muscle damage, muscle soreness, and reduction in voluntary force production (7). Some of these effects may be attenuated after repeated exposure (67).

Recently, Cuthbert et al. (22) explored the related but distinct concepts of microdosing and minimum effective dosing of exercise within team sport contexts as a framework and suggest that lower volumes spread across multiple sessions may minimize interference between strength, endurance, and power sessions within a microcycle. Moreover, according to the analysis by Androulakis-Korakakis et al. (2), as little as 1 set per exercise performed 3 times per week between 70–85% 1 RM can improve strength in trained athletes. The minimum effective dose may, therefore, be useful for practitioners to default to during periods when accumulated training volumes are deemed very high or in periods where RT adaptations are deemed less important than endurance or skills-based training. In general, moderate RT volumes in the region of 12 weekly sets per muscle group/movement pattern (90) could be prescribed to increase the rate of RT adaptation while minimizing the acute recovery demands.

Practical solutions that autoregulate the volume of RT could potentially impact the fatigue indicators on CT. Because the amount of volume required for adaptation can be highly variable (30,89), it can be speculated that by monitoring and individualized RT volumes, we could improve the stimulus-to-fatigue ratio, favoring less fatigue and enhancing readiness for subsequent training sessions (74). The velocity of movement and session density are also considerable "volume"-related variables, which provide some trade-offs (75). The use of VL thresholds is an emerging strategy used in resistance exercise, which involves terminating the set once the bar speed reaches a predetermined reduction in velocity (87), which serves the purpose of keeping a set sufficiently stimulating without reaching momentary failure. Higher VL can impair rapid force production and induce greater fatigue in the short term, which may interfere with subsequent "endurance" or sports-specific sessions (75,86). With this in mind, velocity-based training may have practical significance in helping to limit training volume to the amount necessary to stimulate improvements. Dose-response relationships with hypertrophy (92) and strength (27) have been observed in the literature, and higher volumes are typically recommended to maximize adaptations in muscle hypertrophy (27). However, in the context of intermittent field sport athletes, the diminishing returns with higher training volume may not be worth the trade off with additional fatigue incurred. Therefore, a low VL (~10–20%) is recommended as a set completion threshold. This approach can stimulate improvements while reducing recovery time and stress in the following 24–48 hours, compared with a higher velocity threshold. This is achieved by limiting volume load and the number of repetitions closer to muscle failure, which are associated with greater fatigue accumulation and a longer time course of recovery (41,52,74).

From a team sport perspective, we acknowledge the resource issues associated with VL implementation. The use of subjective tools such as rate of perceived exertion (RPE) or its inverse repetitions in reserve (RIR) may provide an alternative method to individualize training volume (58,110). Similar to the velocity stop method, an RPE stop could limit the volume per exercise performed by prescribing a fixed load with an autoregulated number of sets alongside a suggested repetition range and proximity to failure, such as 2–4 sets of 5 repetitions at 85% 1RM, which terminates at RPE 7 or RIR 3. Unlike the velocity stop method, the use of a subjective gauge of proximity to failure on an exercise is limited by experience using RPE and RIR, and resistance training age (35,42). Current research primarily focuses on recreational and powerlifting trainees, leaving a research gap in CT considerations for field sport athletes, where RT is supplementary and not their primary sport training. This scarcity of data limits the confidence in recommending these methods despite emerging evidence of their long-term benefits within RT (35). However, experienced coaches will be capable of deciding if their athletes can successfully follow these prescriptions effectively.

Session order. Given that training schedules for team sports are constrained by competition schedules and limits on training time, athletes can often be required to perform skill-based sessions, resistance exercise and energy system conditioning on the same day (1). The literature has described acute molecular signaling events that serve to underpin training

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adaptations to both resistance and endurance (aerobic/anaerobic conditioning) modalities. Although imprecise at our current level of understanding, in general terms, the peroxisome proliferator-activated receptor-γ coactivator-1α (PGC-1α) signaling pathway is widely accepted as an important mediator of favorable adaptations for endurance performance, such as mitochondrial biogenesis and fatty acid oxidation (5,108). Conversely, the mammalian target of rapamycin complex 1 (mTORC1) regulates a cascade of downstream signaling events associated with stimulation of muscle protein synthesis (25). This has implications for our context because of the so-called "molecular interference hypothesis," which has been reviewed in detail elsewhere (26). Briefly, this phenomenon posits that the molecular cell signaling transduction pathways associated with skeletal muscle hypertrophy may be blunted by those pathways associated with endurance exercise as interpathway crosstalk results in inhibition of mTORC1 signaling and, therefore, the inhibition of muscle protein synthesis (25). This finding is not universal, and the supposed signaling incompatibility may be absent when different modes of energy systems training, such as HIIT, are deployed as the aerobic stimulus concurrently with RT (50).

Although these studies are at the forefront of our understanding regarding the mechanism of response and adaptation to exercise, previous authors have acknowledged the issues of "real world" applicability (51). The discrepancies observed are likely because of heterogeneity in participants, as the acute molecular responses within skeletal muscle to single out of exercise vary in untrained, endurance trained, and resistance trained cohorts (26). Moreover, from a study design perspective, it makes sense to examine divergence, meaning using polar opposite training sessions, as this has the highest likelihood of demonstrating an effect. However, this may further decrease the transferability from laboratory to field, where strength and aerobic sessions are designed with more nuance. Molecular interference may play out over a very long term (i.e., several years) when pursuing the extremes of sports requiring mostly endurance, for example, triathlon, or mostly strength, for example, powerlifting. However, in the context of team sport, athletes are looking to meet or exceed minimum physical and physiological demands at the given competitive level, which simply "permits" expression of skill within the context of the sport. It is important to note that this is speculative, as current state of the interference effect literature research, only covers weeks and months, not several years. Moreover, pioneer researchers on molecular responses to exercise and training have highlighted that there are likely too many gaps in our understanding to make simplistic recommendations on session order at this point (20).

A more actionable avenue for strength and conditioning practitioners is the consideration of the acute fatigue hypothesis, as highlighted by a recent review where "residual fatigue" and substrate depletion may impact back-to-back training sessions (73). A systematic review and meta-analysis of the effects of intrasession order on the interference effect supports the notion that greater outcomes are achieved from a strength-endurance session order (compared with endurance-strength), at least when endurance type exercise is performed at low intensity (24). There is limited research on high(er) intensity endurance exercise in this context. When performed 4 hours before RT, both cycling and running modes of SIT failed to compromise a subsequent RT session in trained rugby players (66). Similarly, in young men, improvements in strength and endurance were unaffected by session order after 9 weeks' CT when HIIT and RT were separated by 3 hours (62). Clearly, further work is needed to elucidate the effects of high-intensity endurance sessions on subsequent RT sessions (and vice versa) in highly trained male and female team sport athletes, who would be expected to have a greater training age and weekly training demands compared with the cohorts described in the literature.

Finally, the molecular interference effect has been mostly observed locally in the lower body because of the relative ease of extracting a muscle sample from the *m. vastus lateralis* (21,107). Similarly, most of the research investigating the effects of intrasession order has looked at the effects of cycling or running on lower limb strength, hypertrophy, and power outcomes. Therefore, same day, and even same-session, upper-body RT with lower limb dominant aerobic exercise (and vice versa, although this is an unlikely scenario for most sports) is a viable strategy to address concerns about cell signaling interference.

Some interference is inevitable when lower limb RT and predominantly lower-body aerobic training are performed in close proximity. This interference may arise from divergent pathways at the molecular level or simply from the carry-over of acute neuromuscular, metabolic, and/or psychological fatigue from one session to the next. Notwithstanding the limitations outlined, the current state of the literature, taken together with the coaches' experience, still permits practitioners to use a pragmatic approach to intrasession order with enough confidence to achieve their programming goals. If, as determined by needs analysis, or indeed a time window is available (off-season/preseason, return from injury, lack of playing time), it may be prudent to prioritize strength before aerobic training where possible. However, if the goal is to maintain strength, size, and power during competition, it is unlikely, given the current body of evidence, that any tangible effects of exercise session order will occur when a pragmatic approach to programming and periodization is adopted.

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NONTRAINING FACTORS

Travel. A less commonly discussed aspect to consider is the travel fatigue experienced by modern athletes, who often have demanding travel schedules both domestically and internationally (48). These travels include high frequency, short-distance trips (<3 hours) and low frequency, long-distance journeys (>3 hours), often spanning multiple time zones (48,105) (Figure 2). The resulting travel fatigue and jet lag can lead to various detrimental effects, such as disrupted sleep patterns, daytime fatigue, and decreased focus and alertness (105). These effects can potentially interfere with neuromuscular recovery and metabolic adaptation, increasing the challenges of managing training loads during congested schedules. These effects can ultimately impair athletic performance and heighten the risk of injury and illness (31,105), which have the potential to interfere with neuromuscular recovery and metabolic adaptation, increasing the challenges of managing training loads during congested schedules. Therefore, practitioners should be cautious and develop evidence-based travel management strategies to lessen the impact on physical performance and on injury/illness.

Travel fatigue and jet lag should not be used interchangeably, as travel fatigue occurs in all traveling athletes and can either be acute after a long travel or cumulative as a result of repetitive travel within a competitive season (85). Travel fatigue is a naturally occurring phenomenon because of traveling conditions, such as prolonged hypoxia and changes in the environmental conditions irrespective of travel direction and distance (82). Jet lag refers to a condition characterized by more severe symptoms, such as indigestion, diarrhea, and sleep disturbance and, unlike travel fatigue, is fully dependent on the travel distance and travel direction (48), while it typically occurs after traveling across 3 or more time zones in a single journey (82). Responses to jet lag are typically a result of desynchronization between the internal circadian system and the time at the new destination (82). This desynchronization may impair recovery from RT or reduce performance during subsequent high-intensity energy systems sessions, compounding the interference effects inherent to CT. However, it needs to be mentioned that responses are different depending on the direction of the flights and the amount of time zones that are crossed (4,82,105).

Overall, both fatigue management and jet lag can significantly affect performance (31,105). Consequently, travel-induced fatigue and jet lag can further exacerbate the challenges of managing training loads during congested schedules. Disrupted sleep and circadian misalignment may impair recovery from RT or reduce performance in subsequent high-intensity energy system conditioning sessions. Thus, it is imperative for practitioners to devise effective strategies aimed at mitigating the adverse effects to the greatest extent possible. Presently, recommendations for management strategies predominantly rely on expert opinions (48,85) and accumulated practical experience (23,69). Given that frequent travel poses a common challenge for team sports athletes, travel fatigue and jet lag should be regarded as additional hurdles that have the potential to exacerbate fatigue induced by training loads, consequently impeding performance.

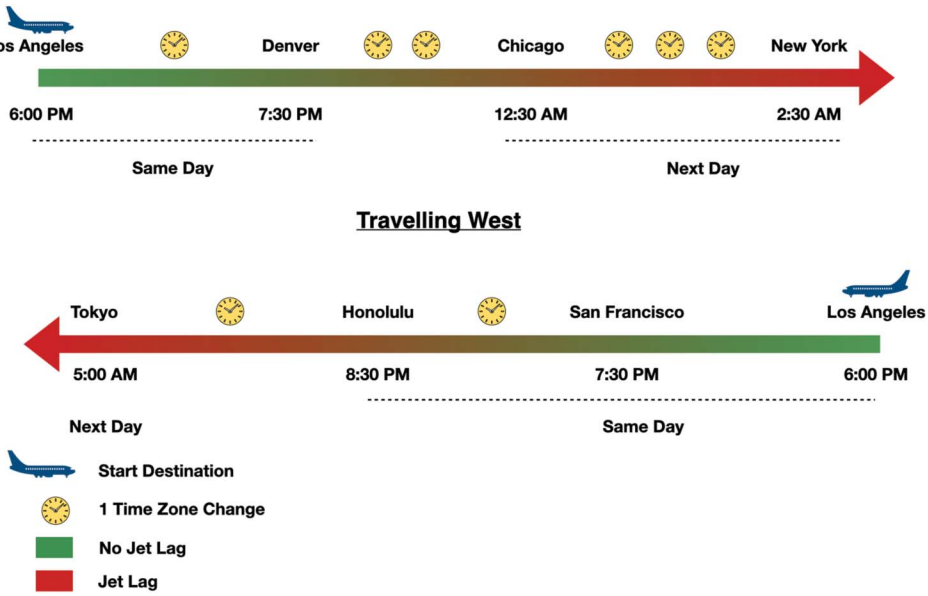

Figure 2. Schematic representation of a travel schedule to different destinations.

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Certainly, the strategies employed would vary depending on whether athletes are traveling westward or eastward, necessitating adjustments to their internal clocks by shifting them either backward or forward, respectively. However, in general, it is recommended that strategies involving light exposure, melatonin administration, and the preservation of sleep patterns can significantly help in the re-entrainment of the circadian rhythm (48).

Generally, light-dark cycle is considered the most important regulator of circadian rhythmicity (95). Therefore, when athletes travel, whether toward the west or the east, adjusting light exposure can serve as a strategic tool for achieving proper phase shifts. According to recommendations, after traveling westward across up to 8 time zones, exposure to natural light typically aids in the desired phase adjustment. Conversely, when athletes travel eastward across a similar number of time zones, it is advised for athletes to avoid morning light, as it can lead to a delay in the desired phase shift (95).

Sleep. Sleep is profoundly influenced by travel (105) and is widely recognized as a critical factor for optimal performance (103). This significance stems from its foundational role in the recovery hierarchy, often depicted as the basis of the recovery pyramid (16). However, despite its importance, it has been shown that athletes suffer from sleep disruptions (103) and decreased sleep quality compared with nonathletes (63). This is surprising considering the importance of sleep on recovery and performance and the potential negative effect of chronic sleep deprivation on performance (40). In particular, it has been shown that chronic sleep deprivation can significantly increase the injury risk (68) and decrease athletic performance and mood (32). On the contrary, the effects of acute sleep loss seem to be mixed (32,40), but there seems to be a clear pattern of worsening sleeping patterns the days leading to competition (29)–potentially because of an increased nervousness–which in combination with travel can further exacerbate the negative effect of sleep loss on performance. The implementation of a structured sleep hygiene program remains one of the most effective strategies for practitioners aiming to enhance both the quality and the quantity of athletes' sleep. Education plays an important role, equipping athletes with evidence-based practices to optimize their sleep. For instance, minimizing blue light exposure within the hour preceding bedtime is widely recognized as a beneficial strategy (40). Moreover, caffeine consumption should be carefully managed; studies indicate that avoiding caffeine intake in the hours leading up to sleep significantly enhances sleep quality. If caffeine consumption is necessary, it is recommended to limit intake to no more than 100 mg and avoid it within a minimum of 4 hours before bedtime (34). Given that sleep disruptions, when combined with the demands of CT and competition, can further complicate the management of training loads and recovery, it seems sleep quality and quantity must be considered when making decisions on training prescription during intensified competition periods. It is important to note that there is no direct research investigating the cumulative effects of sleep disruption and travel demands on CT, highlighting the scope for further research and the need for more context specific, evidence-based approaches.

Nutrition. CT introduces added complexity not only to the training distribution but to the nutritional requirements for athletes as well. This is particularly important in instances where athletes are in negative energy balance, either as part of weight loss strategies (6) or because of an inadequate nutritional understanding (18). This raises the concern regarding muscle mass loss especially when protein intake is suboptimal (6), which can be detrimental for athletic performance (3,18) and skeletal muscle remodeling (17). Therefore, careful consideration of dietary strategies that can support training adaptations while optimizing performance and recovery is necessary (77).

Generally, protein serves as a fundamental element in the body's adaptation process, primarily attributed to its capacity to stimulate muscle protein synthesis (93), thereby contributing significantly to muscle remodeling process (17). Post-exercise protein ingestion can substantially increase contractile myofibrillar proteins' anabolic response (13,17), which is an important step in expediting recovery and alleviating exercise-induced muscle damage (81). Notably, the elevated muscle protein synthesis with amino acid ingestion was present even in the absence of carbohydrate co-ingestion (98), emphasizing the vital role of protein in facilitating muscle growth (98). Hence, the intake of protein emerges as a cornerstone in enhancing exercise-induced adaptations, warranting its prominent inclusion in postexercise regimens to bolster training adaptations. This holds particular significance within CT paradigms, where the accumulation and exacerbation of fatigue, alongside constrained intersession recovery periods, underscore the imperative role of protein in optimizing performance outcomes (77).

Unlike protein, carbohydrates restriction may enhance conditioning training responses, as reflected in a higher activation of molecules involved in regulating aerobic phenotypes, such as mitochondrial biogenesis (13). However, studies have failed to prove that these molecular responses can be translated into improved performance (44,46). In addition, carbohydrate restriction stimulates monophosphate-activated protein kinase, which is well established to inhibit regulatory molecules of muscle protein synthesis, such as the mTORC1, and, therefore, can be detrimental to RT-induced adaptations (76,109). Therefore, given the challenge of limited recovery inherent in

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CT protocols, an emphasis on implementing a nutritionally tailored plan to accelerate recovery is fundamental (77). This strategic approach serves to mitigate interference phenomena and enhance overall performance optimization (77) that will likely support the training recommendations presented in this paper.

Therefore, based on the current findings, it is suggested that after exercise there needs to be a spaced consumption of protein over a 12-hour period equal to 0.25–0.30 $g \cdot kg^{-1} \cdot meal^{-1}$ (64). Nevertheless, considering that athletes' protein requirements are much higher than sedentary individuals, it is advised that athletes should consume ~1.2–1.7 $g \cdot kg^{-1} \cdot d^{-1}$ to support both RT and aerobic training (78). In addition, carbohydrate ingestion at a rate of <1.2 $g \cdot kg^{-1} \cdot h^{-1}$ in combination with protein seems to enhance muscle glycogen resynthesis (15). This is particularly important given the significant muscle glycogen disruption induced by both aerobic (57) and strength components (38). Given that the scope of the paper does not encompass nutritional recommendations, readers are directed to refer to other sources for more details (77).

PRACTICAL APPLICATIONS

The decision tree provided in Figure 3 provides a practical tool for practitioners to use as a guide. We suggest that these are used as a starting point or as troubleshooting for practitioners seeking to make decisions about volume, frequency, intensity, and distribution of resistance and aerobic training during the microcycle. These recommendations are not designed to be prescriptive but rather to simplify programming decisions within the context of the current competition period. We also advise these guidelines can and should be adjusted on the overall programming and periodization strategy, following an evidence-based approach, encompassing practitioner experience and expertise, and athlete preference (28).

CONCLUSION

Congested training/competition schedules induce substantial levels of fatigue that can be detrimental for performance and injury risk (81). Therefore, fatigue management becomes of paramount importance in optimizing the training process (39), but challenges related to the CT training employed in high-intensity intermittent sports need to be considered. Both training and nontraining variables independently and collectively influence the acute fatigue-recovery cycle, which in turn affects subsequent training responses. A working understanding of the influences and interaction of these variables can and should inform decision making in the training process to optimize performance.

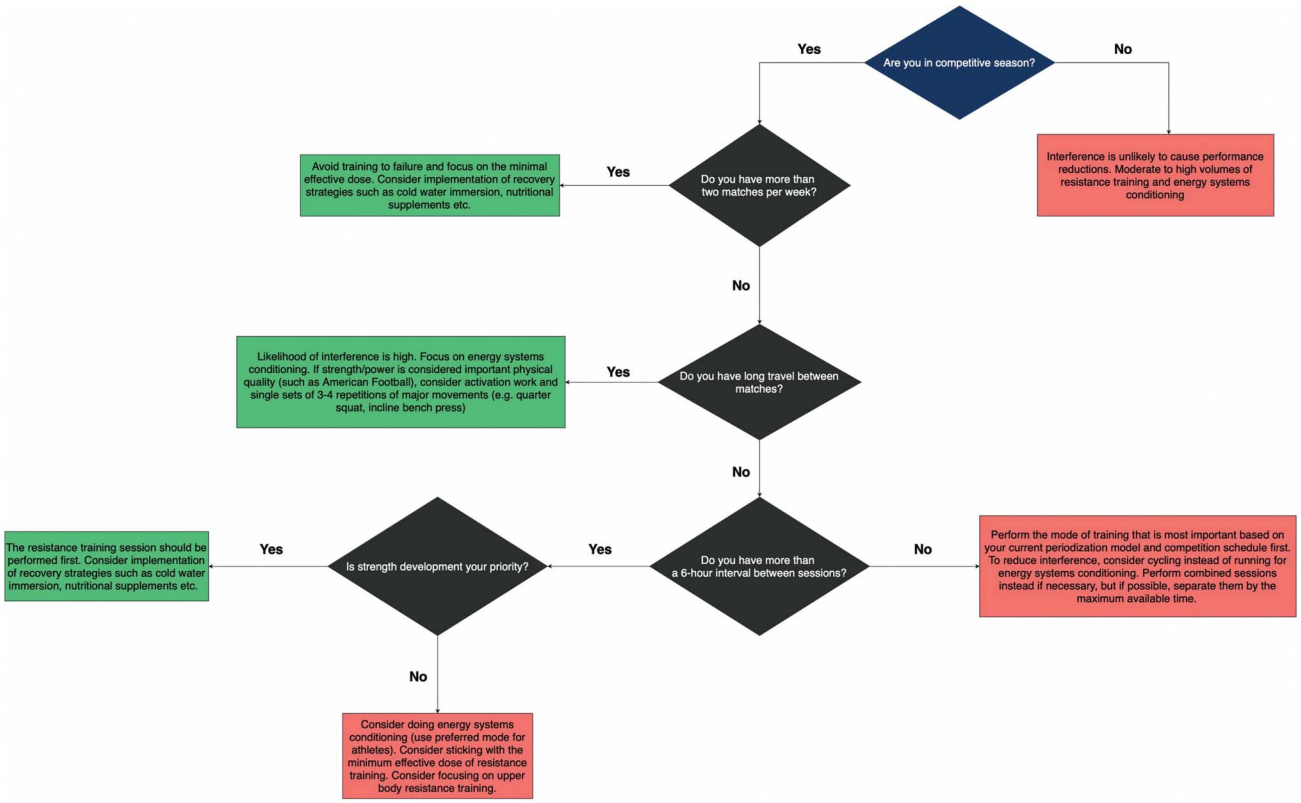

Figure 3. Decision tree to assist manipulation training variables during microcycles depending on the competition load.

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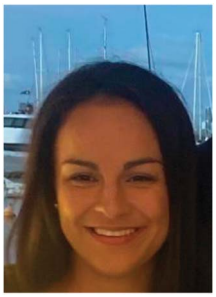

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